# Dodgeball Throw-Attempt Detection

Design document

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**What this is.** The design for a system that turns broadcast dodgeball footage into an attributed timeline of throw attempts and one derived metric, team throw efficiency. It covers the pipeline, the signals, the components, the data and labels, the evaluation, the compute budget and where we expect it to break.
Decisions the build overturned are marked ▹ in place; the measurement and reasoning behind each is in the report.

## Contents

# 1 The event

## 1.1 Why this sport, this event

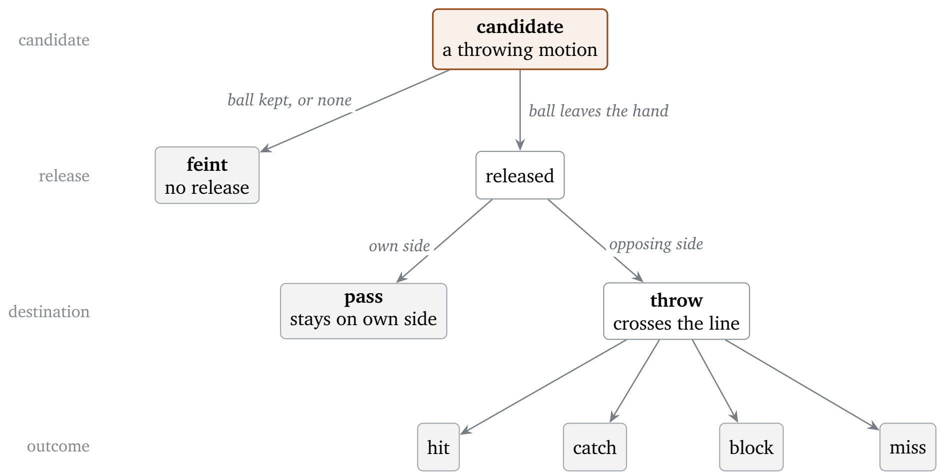
Dodgeball, because it is a less orthodox choice. As far as I can find there is no published vision work on it, one hobbyist post on possession[1] and a 2015 sensor-based paper,[2] so there is no established approach to copy and the design has to come from the sport and the footage. Within it, the candidates below were scored against the brief (hard, fine-grained, temporal, attributable, with a derived metric) and against a ten-hour budget.

| Sport / event | Temporal | Why it was or was not chosen | Verdict |
|---|---|---|---|
| Dodgeball, elimination / re-entry | Weak, a rule consequence | Crisp boundaries and trivial attribution, but it reads as "person crosses a line" | rejected |
| **Dodgeball, throw attempt** | **Strong** | The event does not complete at release, it stays open until the ball resolves, and feints are wind-ups that never release. The ball is only needed near the thrower, so no global ball tracking | **chosen** |
| BJJ, positional change | Strong | Entangled-body pose is unsolved at this budget and labelling needs expertise | rejected |
| Ice hockey, zone entry | Strong | Named in the brief as an example, and puck possession would dominate the budget | rejected |
| Ultimate, turnover | Strong | Viable, but less structured than the throw cascade | fallback |

The throw attempt was chosen for its structured negatives: a pass is identical to a throw up to the last stage, and a feint is identical up to the second.

## 1.2 The event as a cascade

Every labelled thing starts as a *candidate*, a throwing motion, and resolves down a tree. Each level is a separate decision with its own failure mode, so ground truth is labelled at every level and the evaluation reports every level separately (§6) rather than one blended F1.




Terminal boxes in grey. A hit takes one off the side struck; a catch takes the thrower off and returns one to the catcher's side; a block and a miss change nothing. Only hits and catches enter the metric's numerator, only throws its denominator.

**Figure 1:** The event cascade. Four decisions in order, each with its own failure mode and its own score.

| Level | Decision | Where it fails |
|---|---|---|
| Candidate | Is there a throwing motion at all? | Recall ceiling. Whatever is missed here is gone from every level below |
| Release | Did the ball leave the hand? | Occlusion at the release moment: nothing to see, not merely hard to see |
| Destination | Own side (pass) or opposing side (throw)? | Shallow releases near the centre line, lobbed passes |
| Outcome | What happened to the ball? | Graze versus dodge, catch versus drop. This is where annotators will disagree |

[1] A. Hu, 'Leveraging AI Video Frame Recognition to Quantify Ball Possession', *Medium*, December 2025, <https://medium.com/@arthurwang0815/leveraging-ai-image-recognition-to-quantify-ball-possession-e6493d8428e1>, accessed 27 August 2026.

[2] T. Nojima, N. Phuong, T. Kai, T. Sato and H. Koike, 'Augmented dodgeball', in *Proceedings of the 6th Augmented Human International Conference* (AH '15), Singapore, 2015, pp. 137–140.

### 1.3 The event definition

The rules recognise nothing before release: a throw "must leave a player's hand" and the ball is live "once the player is no longer in contact with the ball".[3] A feint is a coaching term, not a rule term, so what counts as one is ours to fix.

| Field | Rule |
|---|---|
| `kind` | `fake` (no release), `pass` (released, ball stays on the thrower's own side), `throw` (released, ball crosses to the opposing side). Mutually exclusive by construction |
| `start` | First frame of the wind-up: the throwing arm moves behind the shoulder line |
| `release` | First frame the ball is no longer in contact with the hand |
| `end` | The first contact that settles the ball: an opponent, a catch, a block, the floor or the far boundary |
| `actor` | Thrower, as team (required) plus player identity where the track is stable |
| `outcome` | `hit`, `catch`, `block`, `miss`, or `unresolved` (occluded or off-frame) |
| `target` | The player the ball reached, for hit / catch / block. Required, because a catch eliminates the *thrower* while a hit eliminates the *target* |
| `confidence` | Reported separately for (a) that a throw occurred, (b) attribution, (c) outcome |

▹ **Changed in the build.** *Not built. Every decision is a threshold on a named quantity and that quantity is written to the timeline (wind-up score, ball mass at the wrist, departure distance, how long the count step held), so a confidence would be a mapping from those margins; it was not done in the budget. Report §2.*

Two things follow from the rules. A candidate needs no ball: an empty-handed wind-up is still a move meant to draw the opponent, and the same motion to a stage that looks for motions. Classification is by destination, not intent: an errant pass that crosses and connects is a throw with a hit, since a live ball eliminates whoever it reaches. The ruleset agrees ("passing throws and plays are not deemed invalid throws, if the ball does not cross into the opponent team's fair territory")[4] and it is the only workable rule, because destination is observable and intent is not. A release whose destination was never seen carries no `kind` at all and is reported separately.

### 1.4 Game state as a free consistency check

Only a throw resolving changes the on-court count: a hit takes one off the side struck, a catch takes the thrower off and returns one to the catcher's side, a miss and a block change nothing. Folding the timeline forward gives the count at every frame, and it has to stay legal. On labels, a violation (seven a side, an eliminated player throwing) locates a bad label without a second annotator. On predictions, the same fold flags impossible sequences as false positives.

### 1.5 The derived metric

**Team throw efficiency** = eliminations ÷ throw attempts, per team per set, split by solo versus coordinated attacks (two or more same-team releases within 300 ms). It is dodgeball's shooting percentage, and the split asks whether attacking together converts more often or just spends more balls.

Its error is the cascade's. Passes and feints sit outside the denominator, so every pass misread as a throw understates efficiency and every feint read as a throw dilutes it. The numerator depends on outcome accuracy. The pass term is the awkward one: a pass causes no elimination and so leaves no trace in the game-state fold, so direction is the only evidence there is.

## 2 Pipeline

Two halves. The first sets up the context every later decision needs (where the court is, who is on it, when the set is live) and runs once per clip. The second is the event cascade itself, one decision per level of the tree in §1.2.

### 2.1 Setting up the context

**Court.** Basic HSV methods pull the lines out of a median background frame and we fit a homography to those. That gives us metres, but the thing it is mostly for is deciding who is in play: anyone whose feet are not on the court is not a player.

**Pose.** YOLO11x-pose over every frame, once, written to disk. Both the pipeline and the labelling tool read the same run, so "no skeleton here" means the same thing to both.

**Set start and end.** We detect the start of each set by looking for the balls lined up on the centre line and a whistle while they are there, confirmed by both teams breaking for the balls. The end is the last elimination. Throws outside a set do not count and the metric is per set.

**Players.** Players get a team from their kit colour and an identity from ByteTrack, with a jersey number read off the best crops where the budget allows. This is what lets us say which side a throw came from and keep a count of

[3] World Dodgeball Federation, *WDBF Dodgeball Rules 2024* (2024), rule 15.2, <`https://worlddodgeballfederation.com/wdbf-content/uploads/2024/03/WDBF-Dodgeball-Rules-2024.pdf`>, accessed 27 August 2026. Quoted by rule number in `docs/reference/wdbf-rules.md`.

[4] WDBF, *Rules*, 16.2.

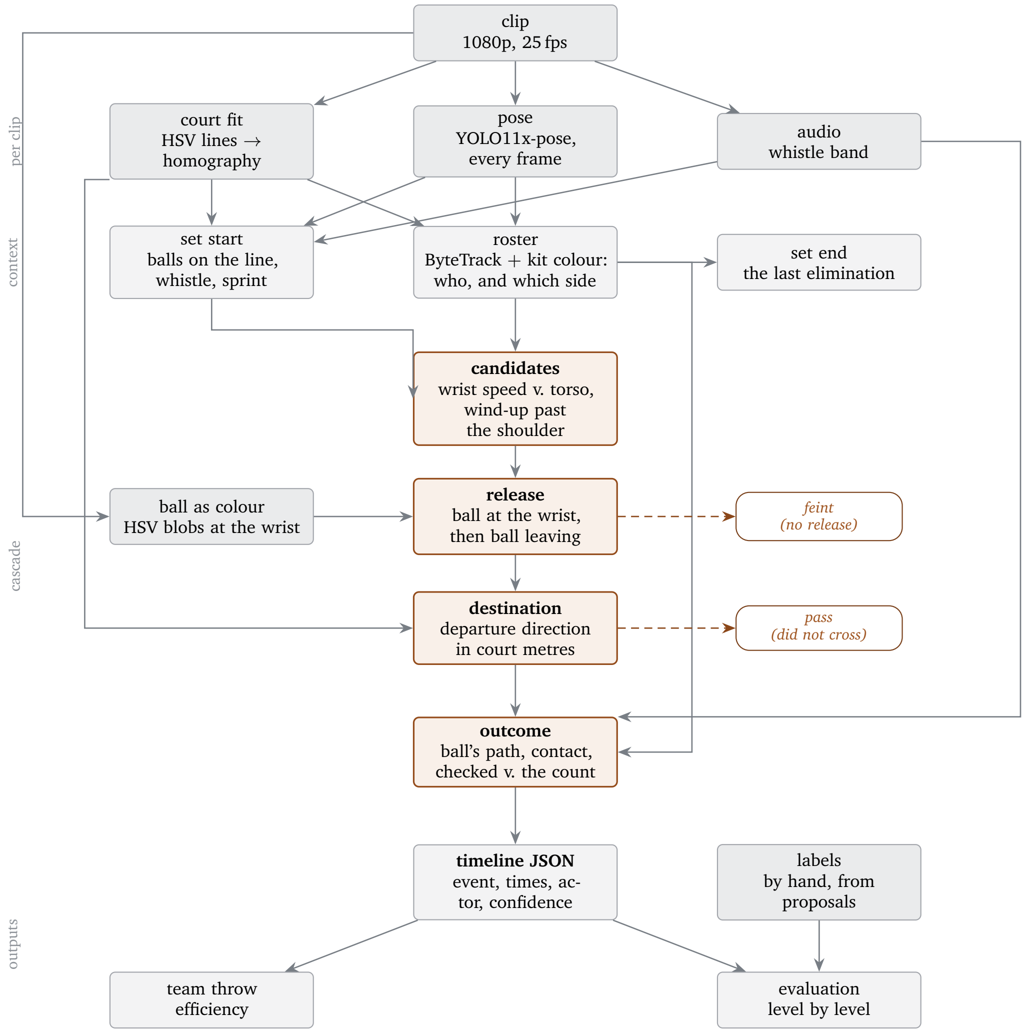



**Figure 2:** Footage in, timeline and metric out, as designed. The four orange stages are the cascade and the dashed exits are where a candidate stops being a throw. Each stage writes one file and the next reads it.

who is still on court.

### 2.2 The event cascade

**Candidates.** Wrist speed, to find when something happens at all. The rest of the cascade works out what that something was.

**Release.** HSV extraction again, this time to check for a ball in the thrower's hands. If the ball is still there afterwards, or was never there, it was a feint. If we can see the ball leaving, it is a throw.

**Destination.** A ball that stays on the thrower's own side is a pass; the court homography says which side it went to.

**Outcome.** For throws at the opposing team we want hit, miss, catch or block. A ball that flies uninterrupted to the floor or the boundary is a miss. Hit, catch and block need other signals on top of the ball's short trajectory: does a player leave the court after it (hit), does a player rejoin (catch), does the ball make contact and nothing changes (block). The on-court count is the check on all of it, since only a throw resolving can move it.

▷ **Changed in the build.** *The outcome is read from the count, not the ball. At 25 fps and twenty pixels the frame where the ball meets someone is the frame the trajectory is lost, so a dodge and a hit look the same; a persistent step in a side's on-court count is the outcome, attributed to the last throw at that side. Blocks move no count and are not claimed. Report §4.*

**Possession** in the team sense does not exist in dodgeball, both sides hold balls at once. Per player it does, and the release check covers the one case that matters: is there a ball in this hand at the wind-up. Tracking it beyond that is possible with what is here, but out of scope.

### 2.3 The stages

| Stage | Writes | What it decides |
|---|---|---|
| `fit_court` | `data/court/` | Homography between image and court metres, the centre line, the margin band, and a scale model for normalising features by depth |

| | | |
|---|---|---|
| `precompute_pose` | `data/pose/` | Every person, every frame, once per clip. Shared by the pipeline and the labelling tool so both see the same detections |
| `detect_set_start` | `data/sets/` | Where each set begins: balls on the centre line arm a window, a whistle inside it is the start, the teams breaking for the balls confirms it |
| `identify_players` | `data/roster/` | Who is a player and which side, from tracks and kit colour |
| `detect_set_end` | `data/sets/` | Where each set ends, the last elimination |
| `detect_candidates` | `data/candidates/` | Throwing motions, proposed loose |
| `detect_events` | `data/timeline/` | The cascade: released or not, pass or throw, and the outcome |
| `evaluate` | `output/` | Every level scored against the labels, plus the metric |

Each stage is its own script and writes its own file, so any stage can be rerun or inspected on its own, and a degraded copy of the clip for the stress test is simply a new clip with its own files. `scripts/run.py` runs the stages in the only order that works and stops at the first failure.

### 2.4 Two contracts every stage shares

- **The clip hash.** Every derived file records the SHA-256 of the clip it was computed from and every reader refuses a mismatch. Frame indices are the only thing tying calibration, detections, labels and timeline together, and a re-encode shifts them without changing a filename.
- **The venue is a file.** What is assumed about *this* venue and *this* footage, the ball's colour window, the court's dimensions, the kit colours, lives in one config file and nowhere else. Algorithm tunings stay out of it, as named constants beside the code that owns them. A second venue should be a config change plus a retune, not a rewrite.

## 3 Signal strategy

### 3.1 The signals considered

| Signal | What it gives | Where it fails | Role |
|---|---|---|---|
| Pose sequence (wrist, elbow, shoulder over ∼0.5 s) | Wind-up onset, throwing arm, the candidate itself | Far-court players at half the pixel height, occlusion by teammates, side-on arm ambiguity | **primary** for candidates |
| Pose at and after the peak (follow-through, arm extension, speed decay) | Possibly feint versus throw | Unknown. A feint is the same motion with the ball kept, so this may carry nothing | to test |
| Ball colour at the wrist | Was there a ball to release | Ball hidden between two bodies, a hue close to the kit | **primary** for release |
| Ball seen leaving | Release, and the first metre of flight | Motion blur, a second ball crossing the wrist | **primary** for release |
| Departure direction in court metres v. the centre line | Pass or throw | Shallow releases near the line, lobbed passes | **primary** for destination |
| Short post-release trajectory | Direction, target side, outcome | Crosses other balls, leaves frame | **primary** for outcome |
| On-court count per side | Hit, catch, and which side lost someone | Eliminations that are not throw outcomes, tracker dropouts | check on outcome |
| Court homography | Which side, the far boundary, depth normalisation | Pan or zoom, few visible markings | geometry layer |
| Audio, whistle band | Set start, and outcome corroboration | Crowd noise, commentary | set start; fusion ablation |
| Global multi-ball tracking | Ball possession counts | Six near-identical balls in constant hand-off | **rejected** |

▷ **Changed in the build.** *Two of these did not hold. Pose at the peak does not separate a feint from a throw, every feature tried sits near chance, so the ball alone decides release. The whistle band fires far too often during play to mean anything without other signals, so audio is used only at the set start, where the ball line-up gives it a window. Report §3.*

### 3.2 Primary approach

Pose finds the motion: the feasibility check (§5.2) says the arm keypoints are clean at near-court scale and coarse but present at far-court scale. The ball decides release, because a feint is a throw with the ball kept and whatever separates them has to be about the ball; follow-through and arm extension get tested because they are cheap, but the design does not depend on them. Direction decides pass versus throw, because a pass leaves no trace in game state and where the ball went is the only evidence; the court homography turns a departure vector in pixels into a side of the centre line.

▷ **Changed in the build.** *Direction is taken in the image, not in court metres, plus contact with a player's box where the trajectory reaches one. The floor homography cannot place a ball in the air. Report §3.*

The trajectory decides outcome, checked against the count: the ball after release says which side it went to and roughly where it ended, and what happened to the player it reached says whether that was a hit, a catch or a miss.

### 3.3 Why HSV and not a detector

HSV is cheap, fast, and enough for what the event needs: the ball at the thrower's wrist and along its first metre of flight, both of which colour gives at this camera. A learned detector is the better long-term choice, but with six balls in play it would need fine-tuning and reliable linking, and that is not worth the budget for a proof of concept. The trigger for building one is the colour mask becoming the limit, a venue where the kit shares the ball's hue or a resolution where the ball is a few pixels. The same reasoning rules out global multi-ball tracking.

## 4 Models and methods

| Job | Chosen | Trade-off, and what was rejected |
|---|---|---|
| Person detection + keypoints | Ultralytics YOLO11x-pose,[5] 1920 px, every frame | The largest pose model at full input resolution, because far-court players are half the height of near ones and the arm keypoints are the whole signal. This is the pipeline's entire compute budget (§7). Smaller variants and 640 px inference are the fallback if it is too slow, at the far court's expense |
| Tracking | ByteTrack,[6] via Ultralytics | Off the shelf and good enough over precomputed detections. Fragments are expected and are what kit colour and, if time allows, jersey numbers repair. Re-identification embeddings are not worth it for twelve players in two kits |
| Team and identity | Kit colour for team; EasyOCR[7] on jersey crops for player identity, budget permitting | Team is required for the metric and kit colour is reliable at this camera. Player identity is reported where a track is stable and a number is readable |
| Court geometry | Median background plate, HSV line extraction, homography to metres, with a held-out check on markings not used in the fit | A known method, and quicker to run than hand-labelling the corners would have been; the venue is ideal for it, one fixed camera and clean markings on a plain floor. It also gives metres, a scale model and a drift check that a hand-drawn polygon would not. Rejected: inference on rectified imagery, since a floor homography stretches upright bodies into shapes no pose model has seen |
| Ball | HSV blob detector, ball-sized components, hue floor set above the red kit | §3. Cheap, and it holds at 20 px where a stock detector drops out on blur. Cannot see a ball against the red team at the far baseline |
| Wind-up | Wrist speed relative to torso, normalised by depth, peak-picking with a score floor | Loose, because a missed candidate is unrecoverable and a rejection costs one keypress in the labelling tool. A learned temporal model is rejected on data volume, sixty events is not a training set |
| Temporal logic | Explicit gates and named constants | Every decision is a threshold on a measured quantity with the reason for its value beside it. Rejected: a fused score, which blends four independent failure modes into one number |

**Nothing is trained.** Every component is pretrained or analytic. With a few dozen labelled events any learned event model would be fitting the evaluation set. Where training would first pay is in §8.

## 5 Data and labels

### 5.1 Footage

Two sources were compared.

| | WDBF 2014 men's final, Canada v. USA, 2nd half[8] | Sky Zone Ultimate Dodgeball 2014 |
|---|---|---|
| Camera | One fixed elevated end-on camera for the whole half | Produced broadcast: cuts every few seconds, replays, commentators |
| Format | 25 fps, 1920×1080 | 30 fps, 1920×1080 |
| Game | WDBF foam, 6-a-side, court fully in frame | Trampoline dodgeball, players airborne much of the time |
| Verdict | **primary** | Rejected. Cuts, re-identification across shots and replay de-duplication would swallow the budget, and "three minutes continuous" is murky when continuity is cut every few seconds |

The evaluation clip is 6:00 to 9:30 of the half: one complete set from the opening rush to the next, 3.5 minutes. All twelve players start on court, so the throw rate is at its highest. It has about 18 seconds of pre-set at the start and a partial lens obstruction part-way through. The following set is the held-out segment.

[5] G. Jocher, J. Qiu and A. Chaurasia, *Ultralytics YOLO* (version 8.4), software, 2024, <https://github.com/ultralytics/ultralytics>, accessed 27 August 2026.
[6] Y. Zhang, P. Sun, Y. Jiang, D. Yu, F. Weng, Z. Yuan, P. Luo, W. Liu and X. Wang, 'ByteTrack: Multi-Object Tracking by Associating Every Detection Box', in *Computer Vision – ECCV 2022*, Lecture Notes in Computer Science 13682 (Cham: Springer, 2022), pp. 1–21.
[7] JaidedAI, *EasyOCR* (version 1.7), software, <https://github.com/JaidedAI/EasyOCR>, accessed 27 August 2026; its detector is Y. Baek, B. Lee, D. Han, S. Yun and H. Lee, 'Character Region Awareness for Text Detection', in *2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition* (Long Beach, 2019), pp. 9357–9366.
[8] World Dodgeball Federation, *2014 World Dodgeball Championship, Men's Final: Canada v USA* [video], YouTube, <https://www.youtube.com/watch?v=Spu6OlAZHUo>, accessed 27 August 2026.

Footage is never committed, for licensing and size. Two scripts obtain it and cut the clip, re-encoding so that frame indices are exact.

### 5.2 Feasibility check

Ten seconds of the clip were run through the intended pose model before the design was fixed.

| Observation | Measured | Consequence |
|---|---|---|
| People per frame | 38.5 mean, against 12 on court | Sidelines, bench, referees and crowd. The court polygon filters them, and the fixed camera means no per-frame work |
| Near-team box height | 280 px median | Full skeletons, clean arm keypoints, wind-ups clearly resolvable |
| Far-team box height | 150 px median, 91 px at p10 | Skeletons present, arm keypoints coarse. Far-court detection will be weaker and is reported separately |
| Ball in hand, near court | 40–50 px, orange on red and grey | Unmistakable unless between two bodies |
| Ball in flight, far court | 15–20 px, still an obvious orange blob | Colour carries a lot. An HSV detector is viable |

### 5.3 Labels

**Ground truth.** No public dodgeball event dataset exists, so it is labelled here. The target is at least twenty events; the estimate for one set is 50 to 70 throws plus 10 to 20 feints at about 45 seconds each, around an hour with a purpose-built tool. How often teams pass is unmeasured, and it sets both the labelling cost and how much the denominator depends on pass rejection.

**The labelling tool.** We will build our own labeller, since the right tool makes the work quicker and less error prone. It will show the pose run over the video with a key for each player on the field, so the thrower and target attach to an event in one press, and every action is a keybinding. It will be bootstrapped with a simple wrist-speed model tuned for recall over precision; I then review each proposal and correct the labels, scrubbing the clip for anything the proposer missed. The cost of bootstrapping is that candidate recall is measured against a truth set the candidate stage seeded, so it reads as the precision of a recall-tuned stage.

**Anchor on the release frame.** It is the crispest moment and the one the temporal tolerance is measured against.

**Player boxes.** A player is stored as a box in source pixels on the frame it was placed, the thrower at release and the target at the end. Evaluation matches boxes to any detector's output, so nothing the detector does later can change what a label means.

**Separating label uncertainty from model error.** Each event records `release_visible`, `outcome_visible`, `ref_signal` (the closest thing to external truth on a contested hit), an `uncertain` flag and a note. These are the error budget's label term.

**Double-label a subset.** About twenty events, second pass done blind, to measure agreement on the release frame and the outcome. This is the label-uncertainty term in the error budget.

**Not labelled.** Non-throw negatives (hand-offs, pickups, retrieval rolls); every false positive gets a cause category during analysis instead. Eliminations follow from outcome and target.

### 5.4 The annotation rule handed to another person

`docs/labeling-guide.md`, written to stand without this document. It fixes the release-frame anchor, where thrower and target are clicked, team from the court half, `kind` by destination, what is not a candidate, and the four ambiguous cases (occluded release, graze versus dodge, crossing balls, unobserved destination) with the flag to set and the fallback to use for each.

## 6 Evaluation

### 6.1 One score per level

Matching is same-frame, same-player: a prediction matches a labelled event when the release frames are within tolerance *and* the thrower boxes overlap.

| Level | Metric and tolerance | What the number misses |
|---|---|---|
| Candidate | Precision / recall / F1 at $\pm 0.25$ s ($\pm 6$ frames at 25 fps) and IoU $\geq 0.5$ on the thrower | Recall is against a truth set this stage seeded (§5.3). Precision is scored against unlabelled negatives, so a wound-up ball the annotator did not call a feint counts against it |
| Boundaries | Mean absolute error on `start` and `end` for matched events | Softer moments than release; expect wide error |
| Attribution | Team accuracy; player accuracy where the track is stable | Team is nearly free at this camera. Player-level attribution only means anything for players whose identity is stable |
| Kind | Feint / pass / throw accuracy on matched events, feints reported separately | With few labelled passes the pass column has a wide interval |

| | | |
|---|---|---|
| Outcome | Accuracy and a confusion matrix over the five classes, on matched throws | Small denominator, and `unresolved` has to be kept apart from wrong |
| Metric | Absolute error in throw efficiency per team | Numerator and denominator errors can cancel. The error budget exists to catch that |

The $\pm 0.25$ s tolerance is about the window a second annotator would place a release in.

### 6.2 Stress

Three conditions, each chosen to break a different thing, with tolerance unchanged. Every pixel-reading stage is rerun on the degraded clip; the court fit, set boundaries and labels are carried over as deterministic transforms of the source, so what is measured is the vision.

| Condition | What it degrades | What we expect it to break |
|---|---|---|
| 480p downscale | Spatial resolution | The ball (8 px is a few pixels of hue) and far-court keypoints |
| Heavy compression (high CRF) | Blocking artefacts | Pose itself, and therefore the wind-up detector's precision |
| Frame dropping (every second frame) | Temporal resolution | The release check, which needs consecutive frames of the ball leaving |

### 6.3 Ablation

One pipeline run scored three times with the later stages withheld: pose only (every proposal is a throw, the simple baseline), plus the release gate (released means throw), plus destination (pass or throw). The candidate matching is identical in every row; what moves is what each stage claims about the same motion. This prices the two structured negatives separately, which matters because they cost the metric differently. A fusion ablation, vision only against vision plus whistle, is the second comparison if audio earns its place.

### 6.4 Error budget

Three terms. **Model**: a paired bootstrap over matched events, giving a 95% band on predicted minus truth efficiency per team. **Labels**: efficiency recomputed with every `uncertain` event decided the other way, plus the double-label agreement. **Sampling**: a Wilson interval on the truth itself. With fifteen throws a side the sampling term will probably dominate, in which case one set cannot measure a team, which is the argument for the pipeline.

## 7 Compute and latency

**Batch.** Pose is essentially the whole cost and everything after it should be seconds.

| Stage | Budget, 3.5 min clip | Basis |
|---|---|---|
| Pose (YOLO11x-pose, 1920 px, half precision) | $\sim$5 min | About 18 fps on a laptop RTX 4080, measured on ten seconds of the clip. Chunked and resumable, since it is the one stage not cheap to rerun |
| Court fit | Seconds, once per clip | A median plate and a line fit |
| Tracking + identity | Under a minute | Tracking over precomputed detections; OCR on selected crops only |
| Candidates, release, destination, outcome | Seconds to a minute | Arithmetic over the pose run, plus one sequential decode of the clip around each proposal for the ball |
| Evaluation, error budget | Seconds | |
| **Training** | **none** | Every component is pretrained or analytic |

**Latency.** As designed the system is batch: a clip in, a timeline out. Near real time is reachable, because nothing in the cascade needs the future beyond a fixed short window: the trajectory is followed a few hundred milliseconds past release, and the count check holds for a second or two. So the floor is a few seconds behind live, per event. Frame-level real time is not reachable and not needed for a metric reported per set.

**Throughput.** Pose is the bottleneck at about 18 fps against 25 fps of footage. Two ways to close the gap, in the order we would try them:

- *Parallelise.* Pose is per frame with no state between frames, so the clip splits into chunks that run on separate GPUs (AWS Batch or similar) and wall-clock falls with the number of workers. No accuracy cost.
- *Speed up one GPU.* TensorRT export first; a smaller pose variant or a lower input resolution last, since both cost accuracy at the far court, which is what full resolution is buying. Half precision is already assumed above: it doubles throughput at this model size for a sub-pixel change in keypoints. Batching does not help, one 1920 px frame already fills the GPU.

## 8 Failure modes

Where we expect the system to break first, and how we would find out.

| Failure | Why | How to test or reduce it |
|---|---|---|
| Feints read as throws | Pose alone cannot tell them apart and the ball check depends on seeing the ball at the wrist | The ablation prices it directly. If the release gate does not buy precision, the ball detector is the next model |
| Passes read as throws | Shallow releases near the centre line, lobbed passes; and no game-state check exists for a pass | Report the pass column on its own. More labelled passes are the only real fix, and the clip may not have many |
| Far-court recall | Half the pixel height, coarse arm keypoints | Report near and far separately. Reducing it means resolution, which means compute |
| Outcome attributed to the wrong throw | Coordinated attacks put several releases within 200 ms and balls cross in flight | Attribute by thrower, never by following the ball. The count check catches an impossible sequence but cannot say which of two throws did it |
| Occluded release | The thrower's hand hidden by a teammate at the moment that matters | Unobservable. Label it as such (`release_visible`) so it lands in the label term, not the model term |
| The ball is a colour | The mask cannot see the ball against a same-hue kit or at a few pixels | The 480p stress condition measures it. That is the trigger for a learned detector |
| Identity | Tracker swaps in pile-ups; unreadable numbers on one kit | Attribute by team, which is what the metric needs. Player-level claims only where a number is read |
| A second venue | Court fit, ball colour and kit colours are all this footage's | Fail loudly: the court fit checks itself against held-out markings, the venue file refuses unknown keys, every derived file refuses a clip it was not computed from |

The biggest structural risk is the outcome. Every other failure degrades gracefully; an outcome on the wrong throw is two errors at once, one invented and one missed, both in the metric's numerator.

## 9 What changed in the build

Three decisions above are marked ▷. The outcome is read from the on-court count rather than the ball's trajectory. Audio is used at the set start only. Pass versus throw is decided by the ball's direction in the image and by contact with a player, not by direction in court metres. Two things were planned and not done: the per-event confidence, and the blind second labelling pass. Each of these has its measurement and its reasoning in the report, and the shipped design of each stage is in `docs/architecture/`.

## Bibliography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Claude Code was used throughout: drafting modules and tests against a design agreed first, running the evaluations, and writing the architecture documentation as each stage shipped. Every label was placed by hand.